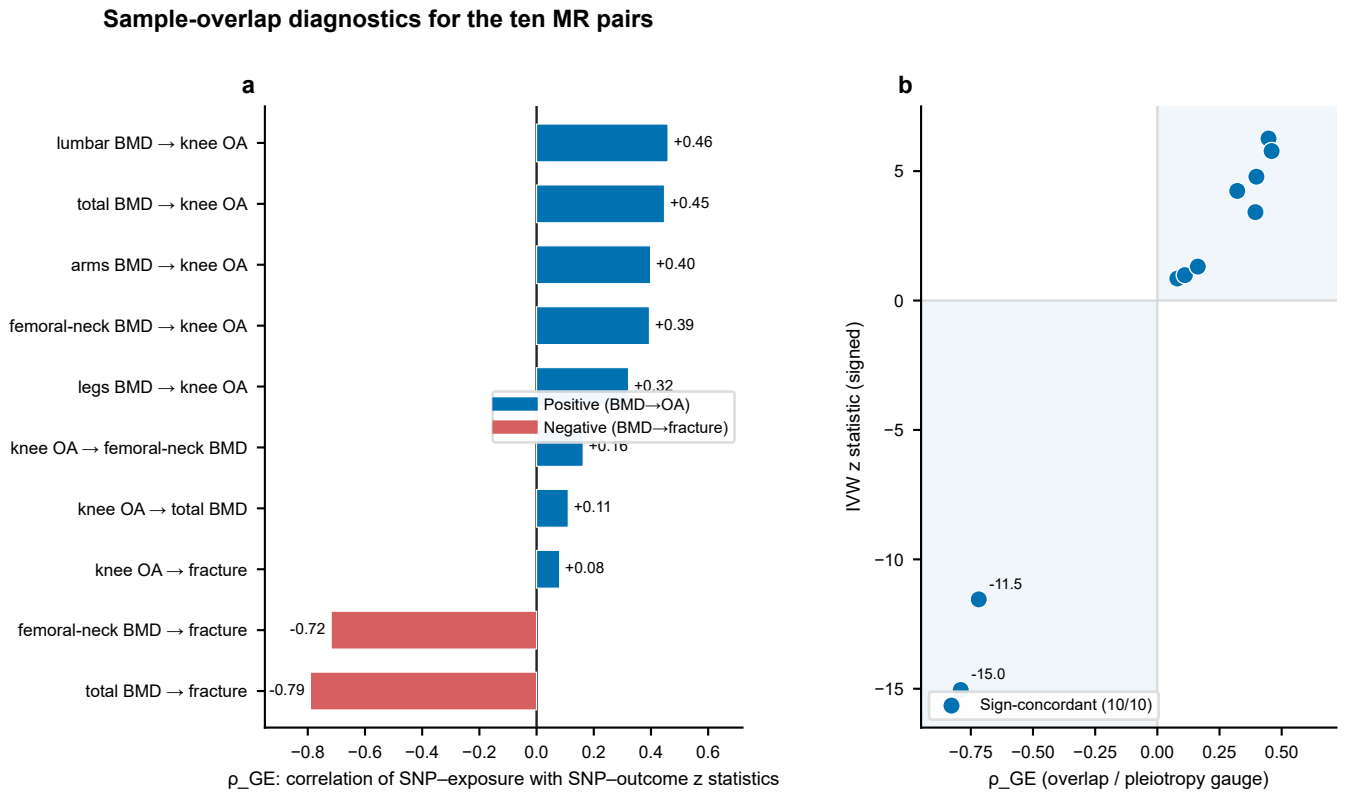
Sample-overlap diagnostics for the ten MR pairs
a
lumbar BMD → knee OA
total BMD → knee OA
arms BMD → knee OA
femoral-neck BMD → knee OA
legs BMD → knee OA
knee OA → femoral-neck BMD
knee OA → total BMD
knee OA → fracture
femoral-neck BMD → fracture
total BMD → fracture
+0.46
+0.45
+0.40
+0.39
+0.32
+0.16
+0.11
+0.08
-0.72
-0.79
Positive (BMD→OA)
Negative (BMD→fracture)
−0.8 −0.6 −0.4 −0.2 0.0 0.2 0.4 0.6
ρ_GE: correlation of SNP–exposure with SNP–outcome z statistics
b
IVW z statistic (signed)
5
0
−5
−10
−15
-11.5
-15.0
Sign-concordant (10/10)
−0.75 −0.50 −0.25 0.00 0.25 0.50
ρ_GE (overlap / pleiotropy gauge)